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Inventor(s)	Ishii; Akira

Sheet feeding device and image forming apparatus including same

Abstract

A sheet feeding device includes a cassette storage part, a sheet feed cassette, a pickup roller, a feed roller, a retard roller, a retaining member, and a pushing mechanism. The pushing mechanism pushes the retaining member so that the retard roller is pushed to contact with the feed roller. The pushing mechanism includes a pushing member that contacts and pushes the retaining member, and a biasing member that biases the pushing member. In a plane perpendicular to a rotation shaft of the retard roller, a pushing direction of the pushing member in a push contact position against the retaining member forms an angle of 45 degrees or less, with a straight line connecting the center of the rotation shaft of the retard roller and the center of a rotation shaft of the feed roller.

Inventors:	Ishii; Akira (Osaka, JP)
Applicant:	KYOCERA Document Solutions Inc. (Osaka, JP)
Family ID:	1000008821624
Assignee:	KYOCERA Document Solutions Inc. (Osaka, JP)
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Primary Examiner: Gokhale; Prasad V

Attorney, Agent or Firm: Stein IP, LLC

Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2023-047619 filed Mar. 24, 2023, the entire contents of which are hereby incorporated by reference.

BACKGROUND

(2) The present disclosure relates to a sheet feeding device and an image forming apparatus including the same.

(3) A conventional sheet feeding device is disclosed in Patent Document 1. This sheet feeding device includes a cassette storage part, a sheet feed cassette, a pickup roller, a feed roller, a retard roller, a retaining member, and a pushing member.

(4) The sheet feed cassette stores sheets and is mounted to the cassette storage part in an attachable and detachable manner. The pickup roller contacts with the sheet stored in the sheet feed cassette so as to send out the sheet. The feed roller conveys the sheet sent out by the pickup roller. The retard roller is pushed to contact with the feed roller to form a nip part, so as to convey the sheet while separating the sheets. If the pickup roller sends out a plurality of sheets at one time, the feed roller and the retard roller feeds only the top sheet by separating the sheets.

(5) The retaining member retains the retard roller so that the retard roller can contact or separate

from the feed roller. The pushing member pushes the retaining member so that the retard roller is pushed to contact with the feed roller. The retaining member is supported in a rockable manner about a rocking axis. Further, in a state where the sheet feed cassette is drawn out from the cassette storage part, the retaining member rocks and moves to a retract position where the retard roller is separated from the feed roller. In addition, in a state where the sheet feed cassette is mounted to the cassette storage part, the retaining member rocks and moves to a push contact position where the pushing member allows the retard roller to be pushed to contact with the feed roller.

(6) When adopting the conventional technique, the pushing of the pushing member against the retaining member may not be stable, and the push contact force of the retard roller against the feed roller may be unstable. If the push contact force of the retard roller against the feed roller is large, double sheet feeding may occur. In addition, if the push contact force of the retard roller against the feed roller is small, a sheet feed error may occur.

SUMMARY

(7) A sheet feeding device according to one aspect of the present disclosure includes a cassette storage part, a sheet feed cassette, a pickup roller, a feed roller, a retard roller, a retaining member, and a pushing mechanism. The sheet feed cassette is mounted to the cassette storage part in an attachable and detachable manner. The pickup roller contacts with a sheet stored in the sheet feed cassette so as to send out the sheet. The feed roller conveys the sheet sent out by the pickup roller. The retard roller is pushed to contact with the feed roller so as to form a nip part, and conveys the sheet while separating the sheets. The retaining member retains the retard roller so that the retard roller can contact or separate from the feed roller. The pushing mechanism pushes the retaining member so that the retard roller is pushed to contact with the feed roller. The retaining member is supported in a rockable manner about a rocking axis extending in parallel to a rotation shaft of the retard roller. In a state where the sheet feed cassette is drawn out from the cassette storage part, the retaining member is positioned at a retract position where the retard roller is separated from the feed roller. In a state where the sheet feed cassette is mounted to the cassette storage part, the retaining member is positioned at a push contact position where the pushing mechanism allows the retard roller to be pushed to contact with the feed roller. The pushing mechanism includes a pushing member that contacts and pushes the retaining member, and a first biasing member that biases the pushing member. In a plane perpendicular to the rotation shaft of the retard roller, a pushing direction of the pushing member in the push contact position against the retaining member forms an angle of 45 degrees or less, with a straight line connecting the center of the rotation shaft of the retard roller and the center of a rotation shaft of the feed roller.

(8) Other objects of the present disclosure and specific advantages obtained by the present disclosure will become more apparent from the description of the embodiment given below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view illustrating an overall structure of an image forming apparatus **100** including a sheet feeding device **50** according to one embodiment of the present disclosure.

(2) FIG. 2 is a schematic cross-sectional view illustrating a structure of the sheet feeding device **50** and its vicinity according to one embodiment of the present disclosure.

(3) FIG. 3 is an exploded perspective view illustrating a structure of a retard roller **32**, a pushing mechanism **60**, and their vicinity of the sheet feeding device **50** according to one embodiment of the present disclosure.

(4) FIG. 4 is an explanatory diagram for describing an operation of the pushing mechanism **60** of the sheet feeding device **50** according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

(5) Hereinafter, an embodiment of the present disclosure is described with reference to the drawings. FIG. 1 is a cross-sectional view illustrating an overall structure of an image forming apparatus **100** including a sheet feeding device **50** of the present disclosure.

(6) The image forming apparatus **100** is a tandem type color copier, and four image forming units Pa, Pb, Pc, and Pd are disposed in a main body of the image forming apparatus **100**, in order from the left in FIG. 1. The image forming units Pa to Pd are disposed corresponding to four images of different colors (yellow, magenta, cyan, and black), and they sequentially form yellow, magenta, cyan, and black images by steps of charging, exposing, developing, and transferring.

(7) The image forming units Pa to Pd respectively include photosensitive drums **1a**, **1b**, **1c**, and **1d** for carrying individual color visual images (toner images). In addition, an intermediate transfer belt **8** that rotates in a counterclockwise direction is disposed adjacent to the image forming units Pa to Pd. The toner images formed respectively on the photosensitive drums **1a** to **1d** are sequentially transferred and overlaid onto the intermediate transfer belt **8** that moves while contacting the photosensitive drums **1a** to **1d**.

(8) After that, the toner images are transferred onto a sheet **26** such as a paper sheet by action of a secondary transfer roller **9**. Further, the toner images are fixed onto the sheet **26** in a fixing device **7**, and then the sheet is discharged from the main body of the image forming apparatus **100**. The photosensitive drums **1a** to **1d** rotate in a clockwise direction in FIG. 1, while image forming processes are performed on the photosensitive drums **1a** to **1d**.

(9) A cassette storage part **101** is disposed in a lower part of the main body of the image forming apparatus **100**. A sheet feed cassette **10** that stores the sheets **26** is mounted to the cassette storage part **101** in an attachable and detachable manner. The sheet feed cassette **10** is attached or detached from the front of the main body of the image forming apparatus **100** (the front side of the paper of FIG. 1).

(10) The sheets **26** are loaded on a sheet loading plate **28** of the sheet feed cassette **10**. A pickup roller **29** is pushed to contact with the top surface of the sheet **26** at a predetermined pressure. When the pickup roller **29** is rotated, feeding of the sheet **26** is started. Out of the fed sheets **26**, the top sheet **26** is separated by a feed roller pair **30** and is conveyed to a sheet conveying path **11**. The sheet **26** after passing the sheet conveying path **11** reaches a registration roller pair **14**, and is conveyed to a nip part between the secondary transfer roller **9** and a drive roller **13** of the intermediate transfer belt **8**, in synchronization with image formation timing.

(11) A dielectric resin sheet is used for the intermediate transfer belt **8**, and a seamless belt is mainly used for the same. In addition, on a downstream side of the secondary transfer roller **9** in the moving direction of the intermediate transfer belt **8**, there is disposed a cleaning blade **17** for removing toner remaining on a surface of the intermediate transfer belt **8**.

(12) An image reading unit **20** is disposed in an upper part of the main body of the image forming apparatus **100**, and reads an original image so as to convert it into image data when performing copying.

(13) Next, the image forming units Pa to Pd are described. Around and below the photosensitive drums **1a** to **1d**, there are disposed charging devices **2a**, **2b**, **2c**, and **2d**, an exposing device **4**, developing devices **3a**, **3b**, **3c**, and **3d**, and cleaning devices **5a**, **5b**, **5c**, and **5d**.

(14) When image data is input from the image reading unit **20**, the charging devices **2a** to **2d** charge surfaces of the photosensitive drums **1a** to **1d** uniformly first. Next, the exposing device **4** emits light beams so as to form electrostatic latent images on the photosensitive drums **1a** to **1d**, respectively, corresponding to the image data. The developing devices **3a** to **3d** have developing rollers (developer carriers) disposed to face the photosensitive drums **1a** to **1d**, respectively. The developing devices **3a** to **3d** are filled with predetermined amounts of two-component developer including toner of yellow, magenta, cyan, and black colors, respectively. The toner is supplied from the developing rollers onto the photosensitive drums **1a** to **1d**, respectively, and toner images corresponding to the electrostatic latent images are formed.

(15) The toner images on the photosensitive drums **1a** to **1d** are primarily transferred onto the intermediate transfer belt **8**. After that, the toner remaining on the photosensitive drums **1a** to **1d** are removed by the cleaning devices **5a** to **5d**.

(16) The intermediate transfer belt **8** is wrapped around a driven roller **12** and the drive roller **13**. When the drive roller **13** rotates, the intermediate transfer belt **8** starts to rotate in the counterclockwise direction. At this time, the sheet **26** is conveyed from the registration roller pair **14** to a nip part (secondary transfer nip part) between the secondary transfer roller **9** and the intermediate transfer belt **8**, at a predetermined timing. At the nip part, the full color image is secondarily transferred onto the sheet **26**.

(17) The sheet **26** is conveyed to the fixing device **7**, and it is heated and pressed when passing through a nip part (fixing nip part) of a fixing roller pair **15**, so that the toner image is fixed to a surface of the sheet **26** and that the predetermined full color image is formed. After that, the sheet **26** passes a conveying roller pair **16**, and the conveying direction thereof is selected by a conveyance guide member **21** disposed at a branch part of a sheet conveying path **19**. Then, the sheet **26** is discharged onto a discharge tray **18** via a discharge roller pair **24** as it is (or after being sent to a double-sided conveying path **23** and after double-sided copying).

(18) The sheet conveying path **19** is configured to connect to the discharge tray **18** or the double-sided conveying path **23**, on the downstream side of the conveying roller pair **16**.

(19) Next, a structure of the pickup roller **29** and its vicinity is described. FIG. **2** is a schematic cross-sectional view illustrating a structure of the sheet feeding device **50** and its vicinity, in which a cross section perpendicular to a rotation shaft **32a** of a retard roller **32** is illustrated. FIG. **3** is an exploded perspective view illustrating a structure of the retard roller **32**, a pushing mechanism **60**, and their vicinity. FIG. **4** is an explanatory diagram for describing an operation of a retaining member **40**, in which a cross section perpendicular to the rotation shaft **32a** of the retard roller **32** is illustrated. In addition, in FIGS. **2** and **4**, the retard roller **32** is in a push contact position where it is pushed to contact with a feed roller **31**.

(20) The sheet feeding device **50** supplies the sheet **26** to the sheet conveying path **11**, and it includes the cassette storage part **101**, the sheet feed cassette **10**, the pickup roller **29**, the feed roller **31**, the retard roller **32**, a roller retaining member **36**, the retaining member **40**, the pushing mechanism **60**, and a separation mechanism **70**.

(21) The pickup roller **29** contacts with the sheet **26** stored in the sheet feed cassette **10** and sends out the sheet **26**.

(22) The feed roller **31** conveys the sheet **26** sent out by the pickup roller **29**. The pickup roller **29** and the feed roller **31** are retained by the roller retaining member **36** in a rotatable manner. The feed roller **31** is attached to a rotation shaft **31a** connected to a not shown drive motor (drive source), and is rotated by a drive force from the drive motor. Between the feed roller **31** and the pickup roller **29**, there is a drive force transmission gear (not shown) that transmits rotation of the feed roller **31** to the pickup roller **29**.

(23) The retard roller **32** is pushed to contact with the feed roller **31** to form a nip part N, so as to convey the sheet **26** while separating the sheets **26**. The retard roller **32** is pushed to contact with the feed roller **31** and is driven to rotate. If a plurality of the sheets **26** are simultaneously fed by the pickup roller **29**, the sheets **26** are separated by the feed roller **31** and the retard roller **32**, and only the top sheet is conveyed to the sheet conveying path **11**.

(24) The retaining member **40** supports the retard roller **32** in a rotatable manner. In this embodiment, the retaining member **40** includes a semi-cylindrical housing part **41** extending in an axial direction of the retard roller **32**, a first pushed piece **43**, and a second pushed piece **44**. The housing part **41** houses the rotation shaft **32a** of the retard roller **32** inside, and supports the rotation shaft **32a** in a rotatable manner.

(25) The rotation shaft **32a** protrudes from the housing part **41** to one side A' in the axial direction. The retard roller **32** is disposed at an end of the rotation shaft **32a** on the one side A' in the axial

direction.

(26) A rocking axis **42** is formed at an end of the housing part **41** on a downstream side B in the sheet conveying direction. The rocking axis **42** extends in parallel to the rotation shaft **32a**. The housing part **41** is supported in a rockable manner, so that an end thereof on an upstream side B' in the sheet conveying direction can move in an up and down direction about the rocking axis **42**. In other words, the retaining member **40** retains the retard roller **32** so that it can contact or separate from the feed roller **31**. In addition, the retaining member **40** is supported in a rockable manner about the rocking axis **42** extending in parallel to the rotation shaft **32a** of the retard roller **32**. Note that the retard roller **32** is disposed outside the housing part **41** in this embodiment, but it may be disposed inside the housing part **41**.

(27) The first pushed piece **43** protrudes to the upstream side B' in the sheet conveying direction from an outer circumference surface of the housing part **41**, at the end of the housing part **41** on the one side A' in the axial direction (see FIG. 2). The first pushed piece **43** is biased by the pushing mechanism **60** in a direction approaching the feed roller **31** (in an upward direction in this embodiment).

(28) The second pushed piece **44** protrudes to the upstream side B' in the sheet conveying direction from the outer circumference surface of the housing part **41**, at the end of the housing part **41** on the other side A in the axial direction (see FIG. 2). The second pushed piece **44** is biased by the separation mechanism **70** in a direction separating from the feed roller **31** (a downward direction in this embodiment).

(29) The pushing mechanism **60** includes a pushing member **61**, a first biasing member **62**, and a connecting member **63**.

(30) The pushing member **61** is disposed on the lower side of the first pushed piece **43** (the opposite side to the feed roller **31** with respect to the retaining member **40**). The pushing member **61** includes a rotating part **61a**, a pushing piece **61b**, and an engaging piece **61c**.

(31) The rotating part **61a** is supported in a rotatable manner by a rotation shaft **61d** extending in the sheet conveying direction (BB' direction). The pushing piece **61b** and the engaging piece **61c** protrude from the outer circumference surface of the rotating part **61a**.

(32) The pushing piece **61b** and the engaging piece **61c** are disposed with a predetermined distance therebetween in the circumferential direction of the rotation shaft **61d**. The first pushed piece **43** is disposed between the pushing piece **61b** and the engaging piece **61c**. In this case, the pushing piece **61b** is positioned closer to a third direction side C3 than the first pushed piece **43** (on the clockwise direction side viewed from the upstream side B' in the sheet conveying direction). In other words, the first pushed piece **43** faces the pushing piece **61b**. On the other hand, the engaging piece **61c** is positioned closer to a fourth direction side C4 than the first pushed piece **43** (on the counterclockwise direction side viewed from the upstream side B' in the sheet conveying direction).

(33) The first biasing member **62** is constituted of a helical compression spring and extends in parallel to the axial direction of the rotation shaft **32a** (an AA' direction). An end of the first biasing member **62** on the one side A' in the axial direction is fixed to the engaging piece **61c**. An end of the first biasing member **62** on the other side A in the axial direction is connected to the main body of the image forming apparatus **100** other than the sheet feeding device **50**, via the connecting member **63**.

(34) The separation mechanism **70** is disposed in the cassette storage part **101**, and it includes a lever **71** and a second biasing member **72**.

(35) The lever **71** is disposed at the end of the housing part **41** on the other side A in the axial direction, on the upstream side B' in the sheet conveying direction. The lever **71** is disposed outside a sheet passing area. The lever **71** is formed in a plate-like shape. The lever **71** is supported in a rotatable manner by a rotation shaft **74** extending in the up and down direction.

(36) An end **71a** of the lever **71** on the one side A' in the axial direction faces the second pushed

piece **44** in the sheet conveying direction (the BB' direction). An end **71b** of the lever **71** on the other side A in the axial direction faces a side wall **10a** of the sheet feed cassette **10** in the sheet conveying direction (the BB' direction).

(37) The second biasing member **72** is constituted of a helical spring, and is attached to the rotation shaft **74** of the lever **71**. The second biasing member **72** biases the lever **71** in a first direction C1 (the counterclockwise direction viewed from above).

(38) The lever **71** rotates between a working position and a non-working position. Specifically, the lever **71** rotates to be positioned at the working position in a state where the sheet feed cassette **10** is not mounted to the cassette storage part **101**, and positioned at the non-working position in a state where the sheet feed cassette **10** is mounted to the cassette storage part **101**.

(39) In the working position, the lever **71** rotates in the first direction C1 by a biasing force of the second biasing member **72**. In this case, the end **71a** of the lever **71** on the one side A' in the axial direction contacts with the second pushed piece **44**. With the biasing force of the second biasing member **72**, the lever **71** pushes the retaining member **40** so as to allow the retard roller **32** to separate from the feed roller **31**. In this way, the nip between the retard roller **32** and the feed roller **31** is released.

(40) When the lever **71** is at the working position, the retaining member **40** is biased by the lever **71** that is biased by the second biasing member **72**, in the direction where the second pushed piece **44** separates from the feed roller **31**, so as to rock and move to the retract position. In this case, the biasing force of the second biasing member **72** is larger than a biasing force of the first biasing member **62**. In other words, a pushing force of the lever **71** against the second pushed piece **44** is larger than a pushing force of the pushing piece **61b** against the first pushed piece **43**, and the retaining member **40** rocks and moves to the retract position against the biasing force of the first biasing member **62**.

(41) The retaining member **40** is restricted from moving further than the retract position in the direction separating from the feed roller **31**. Therefore, the rocked and moved state at the retract position by the biasing force of the second biasing member **72** is maintained. Thus, the state where the nip between the retard roller **32** and the feed roller **31** is released is maintained.

(42) In other words, in the state where the sheet feed cassette **10** is not mounted to the cassette storage part **101**, the lever **71** is positioned at the working position, and the retaining member **40** is positioned at the retract position.

(43) On the other hand, in the non-working position, the end **71b** of the lever **71** on the other side A in the axial direction contacts with the side wall **10a** of the sheet feed cassette **10**. In this case, the lever **71** is pushed to the sheet feed cassette **10** in a second direction C2 (in the clockwise direction viewed from above), so as to rotate against the biasing force of the second biasing member **72**. In this way, the end **71a** of the lever **71** on the one side A' in the axial direction separates from the retaining member **40**, and pushing of the lever **71** against the retaining member **40** is released.

(44) When the lever **71** is in the non-working position, the retaining member **40** is biased by the pushing piece **61b** that is biased by the first biasing member **62**, in the direction where the first pushed piece **43** approaches the feed roller **31** (in the upward direction in this embodiment), so as to rock and move to the push contact position. In this case, the retard roller **32** is pushed to contact with the feed roller **31** at a predetermined push contact force.

(45) In other words, in the state where the sheet feed cassette **10** is mounted to the cassette storage part **101**, the lever **71** is positioned at the non-working position, and the retaining member **40** is positioned at the push contact position.

(46) As illustrated in FIG. 4, in a plane perpendicular to the rotation shaft **32a** of the retard roller **32**, it is preferred that a pushing direction F1, in which the pushing piece (recess) **61b** in the push contact position pushes the first pushed piece **43** (the retaining member **40**), should form an angle $\theta 1$ of 45 degrees or less with a straight line L1 that connects the center of the rotation shaft **32a** of the retard roller **32** and the center of the rotation shaft **31a** of the feed roller **31**.

(47) When the pushing direction **F1** is parallel to the straight line **L1** (when the angle $\theta 1$ is 0 degrees), the push contact force of the retard roller **32** against the feed roller **31** can be most stabilized. Therefore, when the first pushed piece **43** is pushed by the pushing piece (recess) **61b** so that the angle $\theta 1$ between the pushing direction **F1** and the straight line **L1** becomes close to zero, the biasing force of the first biasing member **62** can be efficiently transmitted to the retaining member **40**. In this way, the push contact force of the retard roller **32** against the feed roller **31** can be stabilized, and occurrence of a sheet conveyance error can be suppressed.

(48) Note that when the pushing direction **F1** is parallel to a tangent direction **F2** at a tangent point **P** of a circle whose center is the rocking axis **42**, the biasing force of the first biasing member **62** can be most efficiently transmitted to the retaining member **40**. For this reason, it is preferred that the pushing direction **F1** should form an angle of 45 degrees or less with the tangent direction **F2** at the tangent point **P** of the circle whose center is the rocking axis **42**.

(49) On the other hand, as the angle $\theta 1$ between the pushing direction **F1** and the straight line **L1** becomes larger, a component of the pushing force of the pushing piece (recess) **61b** in the tangent direction **F2** becomes smaller. In this way, the biasing force of the first biasing member **62** transmitted to the retaining member **40** is decreased. Therefore, if the angle $\theta 1$ is larger than 45 degrees, the push contact force of the retard roller **32** against the feed roller **31** becomes unstable.

(50) Note that the embodiment disclosed this time is an example in every aspect and should not be interpreted as a limitation. The scope of the present disclosure is defined not by the above description of the embodiment but by the claims, and should be understood to include all modifications within meaning and scope equivalent to the claims.

(51) For instance, in the embodiment described above, the attachment and detachment direction of the sheet feed cassette **10** with respect to the main body of the image forming apparatus **100** is perpendicular to the sheet conveying direction (the **BB'** direction) as an example, but the present disclosure is not limited to this. The present disclosure can also be applied to a sheet feed cassette whose attachment and detachment direction with respect to the main body of the image forming apparatus **100** is parallel to the sheet conveying direction (the **BB'** direction). In addition, the first biasing member **62** extends in parallel to the rotation shaft **32a** of the retard roller **32** in this embodiment, but it may extend perpendicularly to the rotation shaft **32a** as long as the rotating part **61a** can be biased and rotated.

Claims

1. A sheet feeding device comprising: a cassette storage part; a sheet feed cassette mounted to the cassette storage part in an attachable and detachable manner; a pickup roller that contacts with a sheet stored in the sheet feed cassette so as to send out the sheet; a feed roller that conveys the sheet sent out by the pickup roller; a retard roller that is pushed to contact with the feed roller so as to form a nip part, and conveys the sheet while separating the sheets; a retaining member that retains the retard roller so that the retard roller can contact or separate from the feed roller; and a pushing mechanism that pushes the retaining member so that the retard roller is pushed to contact with the feed roller, wherein the retaining member is supported in a rockable manner about a rocking axis extending in parallel to a rotation shaft of the retard roller, the retaining member is positioned at a retract position where the retard roller is separated from the feed roller, in a state where the sheet feed cassette is drawn out from the cassette storage part, while the retaining member is positioned at a push contact position where the pushing mechanism allows the retard roller to be pushed to contact with the feed roller, in a state where the sheet feed cassette is mounted to the cassette storage part, the pushing mechanism includes a pushing member that contacts and pushes the retaining member, and a first biasing member that biases the pushing member, and in a plane perpendicular to the rotation shaft of the retard roller, a pushing direction of the pushing member in the push contact position against the retaining member forms an angle of 45 degrees or less, with a

straight line connecting the center of the rotation shaft of the retard roller and the center of a rotation shaft of the feed roller.

2. The sheet feeding device according to claim 1, further comprising a separation mechanism that pushes the retaining member so as to allow the retard roller to separate from the feed roller, wherein the separation mechanism is disposed in the cassette storage part, and includes a lever supported in a rotatable manner corresponding to attachment or detachment of the sheet feed cassette, so as to push the retaining member, and a second biasing member that biases the lever, the lever rotates to be positioned at a working position in a state where the sheet feed cassette is not mounted to the cassette storage part, and positioned at a non-working position in a state where the sheet feed cassette is mounted to the cassette storage part, in the working position, the lever pushes the retaining member so as to allow the retard roller to separate from the feed roller, with a biasing force of the second biasing member, and in the non-working position, the lever is pushed by the sheet feed cassette, so as to separate from the retaining member against the biasing force of the second biasing member, and releases pushing of the retaining member.

3. The sheet feeding device according to claim 2, wherein the pushing member includes a rotating part supported in a rotatable manner by a rotation shaft extending in a sheet conveying direction, and an engaging piece and a pushing piece that protrude from an outer circumference surface of the rotating part and are arranged in a circumferential direction, the first biasing member has an end engaged with the engaging piece, so as to bias and rotate the pushing piece, the retaining member includes a first pushed piece that protrudes in the sheet conveying direction so as to face the pushing piece, and a second pushed piece that faces the lever, when the lever is in the non-working position, the retaining member is biased by the pushing piece that is biased by the first biasing member, in the direction where the first pushed piece approaches the feed roller, so as to rock and move to the push contact position, and when the lever is in the working position, the retaining member is biased by the lever that is biased by the second biasing member, in the direction where the second pushed piece separates from the feed roller, so as to rock and move to the retract position.

4. An image forming apparatus comprising: the sheet feeding device according to claim 1; and an image forming unit that forms an image on a sheet.
